

Stability and Bifurcation of a Simple Food Chain in a Chemostat with Removal Rates

Mathematical Modelling Project Presentation

Nguyen Cong Hung Nguyen Tran Minh Tri Bui Huu Vinh Ngo Quang Duc Pham Chi Bang

School of Information and Communications Technology
Hanoi University of Science and Technology

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Presentation Roadmap

- ➊ Biological motivation
- ➋ Mathematical model
- ➌ Equilibria and stability
- ➍ Numerical evidence
- ➎ Persistence and coexistence
- ➏ Hopf bifurcation
- ➐ Effect of removal rates
- ➑ Main conclusion

Core message

Distinct removal rates change the qualitative long-run behavior of the food chain.

Biological setting

A chemostat continuously adds nutrient and removes culture medium, keeping volume approximately constant.

Food chain



- ▶ S : nutrient
- ▶ x : prey
- ▶ y, z : first and second predators

Main question

- ▶ Which species survive?
- ▶ When is coexistence stable?
- ▶ When do oscillations appear?
- ▶ How do D_1, D_2, D_3 reshape the outcome?

State Variables and Biological Meaning

Symbol	Meaning	Role
$S(t)$	Nutrient concentration	Resource consumed by prey
$x(t)$	Prey population	Consumes nutrient
$y(t)$	First predator	Consumes prey
$z(t)$	Second predator	Consumes first predator
D_i	Removal rates	Species-specific loss rates

Feasible state space

$$S(t), x(t), y(t), z(t) \geq 0.$$

$$\begin{aligned}\frac{dS}{dt} &= 1 - S - f_1(S)x, \\ \frac{dx}{dt} &= x(f_1(S) - D_1) - f_2(x)y, \\ \frac{dy}{dt} &= y(f_2(x) - D_2) - f_3(y)z, \\ \frac{dz}{dt} &= z(f_3(y) - D_3).\end{aligned}$$

- ▶ $1 - S$: nutrient inflow and washout
- ▶ $f_1(S)x$: nutrient consumed by prey
- ▶ D_i : removal of population i
- ▶ f_i : biological response rate

General assumptions

$$f_i(0) = 0, \quad f_i \in C^1, \quad f'_i(u) > 0 \quad (u > 0).$$

Monod-type functions used in simulations

$$f_1(S) = \frac{a_1 S}{b_1 + S}, \quad f_2(x) = \frac{a_2 x}{b_2 + x}, \quad f_3(y) = \frac{a_3 y}{b_3 + y}.$$

- ▶ Increasing response with saturation.
- ▶ Biologically interpretable parameters: a_i controls maximum response, b_i controls half-saturation.

Why Distinct Removal Rates Matter

Adding the four equations gives

$$\frac{d}{dt}(S + x + y + z) = 1 - S - D_1x - D_2y - D_3z.$$

If removal rates were identical

A simpler total-quantity relation could be used to reduce the system.

With distinct removal rates

The right-hand side depends on x, y, z separately. The full four-dimensional system must be studied.

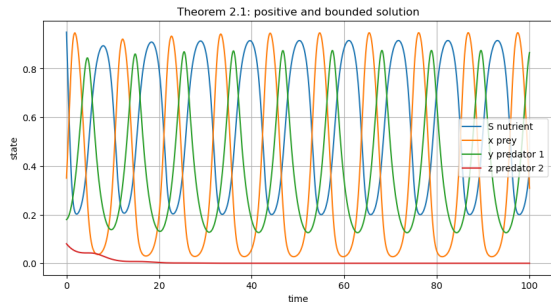
$D_1 \neq D_2 \neq D_3 \Rightarrow$ richer dynamics

- 1 Choose Monod response functions.
- 2 Set parameter values and initial condition.
- 3 Solve ODEs using adaptive Runge-Kutta.
- 4 Discard transient behavior.
- 5 Classify long-run regime.

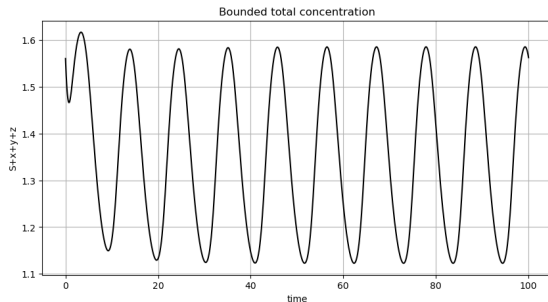
Observed regimes

- ▶ Washout
- ▶ Prey-only survival
- ▶ Boundary coexistence
- ▶ Stable coexistence
- ▶ Oscillatory coexistence

Positivity, Boundedness, and Dissipativity



Positivity and Boundedness



Total Quantity Bounded

- ▶ Solutions remain non-negative and bounded: $\min(S, x, y, z) \approx 2.02 \times 10^{-9}$.
- ▶ Total quantity does not diverge: $1.1228 \leq S + x + y + z \leq 1.6164$.
- ▶ This provides numerical evidence for dissipative behavior and biological feasibility.

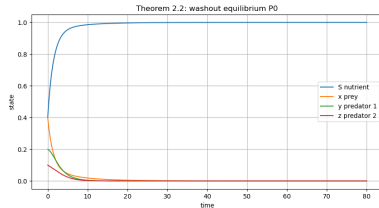
Equilibrium Points

Equilibrium	Form	Biological meaning
P_0	$(1, 0, 0, 0)$	Washout of all organisms
P_1	$(S_1, x_1, 0, 0)$	Prey survives; predators extinct
P_2	$(S_2, x_2, y_2, 0)$	Prey and first predator survive
P_3	(S_3, x_3, y_3, z_3)	Full coexistence

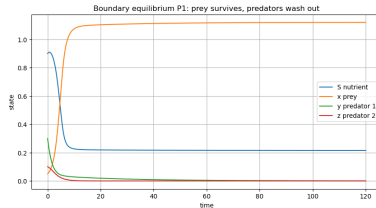
Interpretation

Equilibria are not only algebraic solutions; they represent possible ecological outcomes.

Boundary Regimes: Time-Series and Phase View

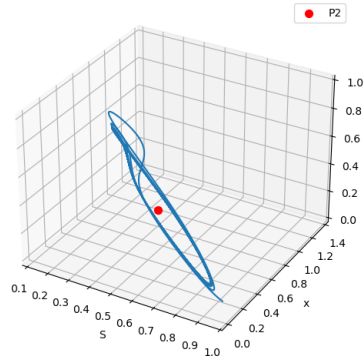


P_0 : washout



P_1 : prey-only

Trajectory approaching P_2 in S-x-y space



P_2 : top predator washed out (phase view)

- ▶ Different parameter conditions remove different trophic levels.
- ▶ Boundary equilibria correspond to real long-run regimes.
- ▶ Trajectories show attraction toward lower-dimensional ecological states.

Linearization and Eigenvalue criterion

$$\max \operatorname{Re}(\lambda_{\text{eig}}) < 0 \quad \Rightarrow \quad \text{locally asymptotically stable.}$$

Eq.	Main condition	Value	Interpretation
P_0	Prey invasion at nutrient-only state	-0.1333	Washout stable
P_1	First-predator invasion at prey-only state	-0.0351	Prey-only survival
P_2	Top-predator invasion $f_3(y_2) - D_3$	-0.1178	z cannot persist

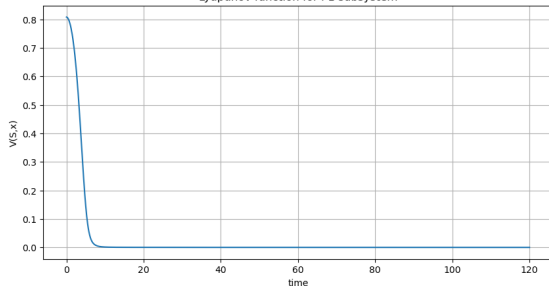
- ▶ Invasion conditions are read from boundary eigenvalues.
- ▶ The P_2 value indicates the top predator cannot invade near that boundary state.

Lyapunov Functions for Global Stability

Prey-only subsystem (P_1)

$$V(S, x) = \frac{1}{2}(S - S_1)^2 + \frac{1}{2}(x - x_1)^2.$$

Lyapunov function for P1 subsystem

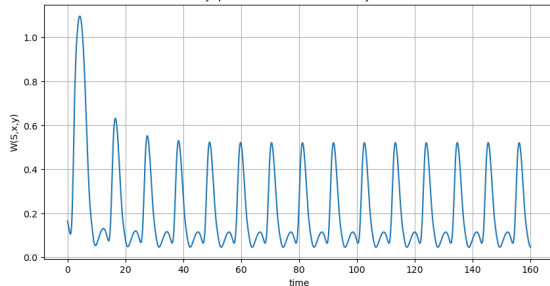


$V(S, x)$ near P_1

Boundary subsystem (P_2)

$$W(S, x, y) = S - S_2 - S_2 \ln \frac{S}{S_2} + \frac{1}{2}(x - x_2)^2 + \frac{1}{2}(y - y_2)^2$$

Lyapunov function for P2 subsystem



$W(S, x, y)$ near P_2

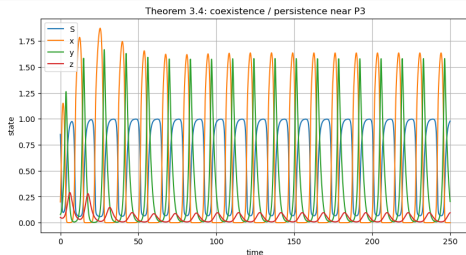
- Profiles decrease toward small values, supporting global stability.

Persistence and Coexistence at P_3

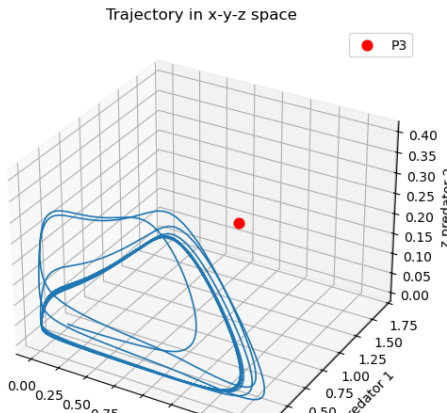
Coexistence equilibrium

$$P_3 = (0.0669, 1.6291, 0.1667, 0.3963).$$

- ▶ All four components are positive, meaning populations stay away from extinction.
- ▶ Behavior is robust across multiple positive initial conditions.



Coexistence time series



Definition

A Hopf bifurcation occurs when a complex conjugate pair of eigenvalues crosses the imaginary axis.

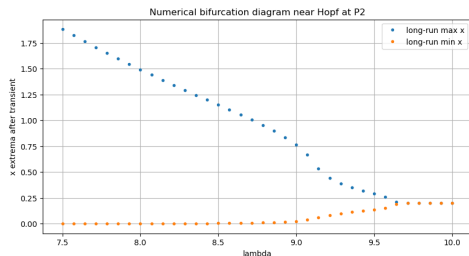
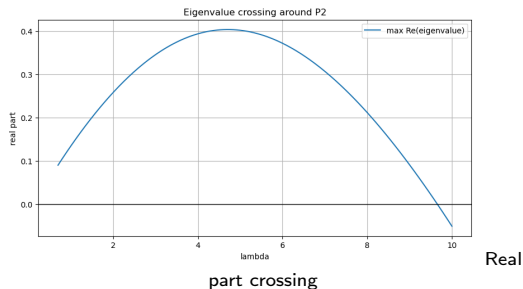
$$\lambda_{\text{eig}}(\lambda) = \alpha(\lambda) \pm i\omega(\lambda), \quad \alpha(\lambda^*) = 0, \quad \omega(\lambda^*) \neq 0.$$

- ▶ Before crossing: equilibrium may be stable.
- ▶ After crossing: oscillations can appear.
- ▶ Ecological meaning: coexistence can be cyclic, not steady.
- ▶ Parameter used: $a_1 = \lambda$.

Hopf Bifurcation Around P_2

Critical point: $\lambda^* \approx 9.6555$, $\lambda_{\text{eig}} \approx \pm 1.6033i$

Complex conjugate pair crosses the imaginary axis.



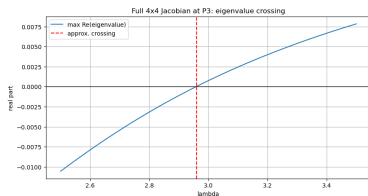
Bifurcation diagram using $x(t)$ extrema

- Split extrema indicate sustained boundary oscillation ($\lambda = 8.5$).
- Coincident extrema indicate convergence to equilibrium ($\lambda = 9.8$).

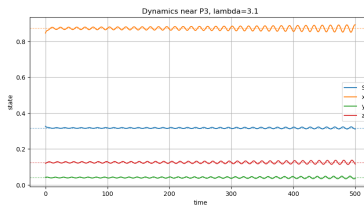
Hopf Bifurcation Around P_3

Critical point: $\lambda^* \approx 2.9582$, $\lambda_{\text{eig}} \approx \pm 0.4722i$

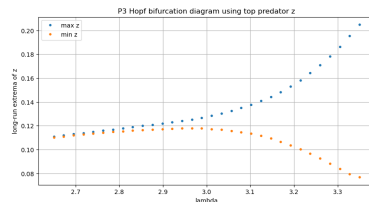
All species can persist, but their densities may oscillate instead of settling to constants.



Real part crossing

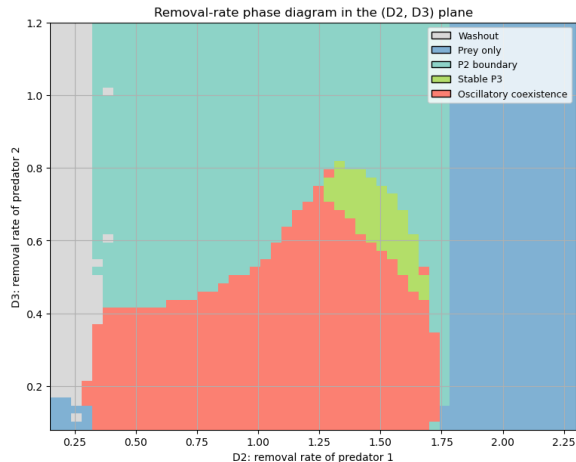


$\lambda = 3.10$: oscillatory side



Top-predator extrema

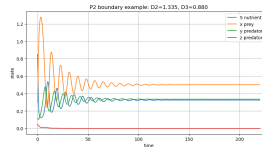
Effect of Removal Rates: Phase Diagram



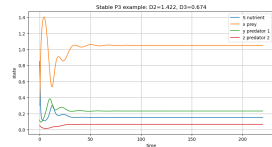
Parameter sweep:

$$D_2 \in [0.15, 2.30], \quad D_3 \in [0.08, 1.20]$$

D_2 , D_3 strongly shape survival and oscillation.



P_2 boundary



Stable P_3

- Large predator removal rates eliminate higher trophic levels.

Main conclusions

- ▶ Positivity and boundedness are numerically supported.
- ▶ Boundary regimes match invasion conditions.
- ▶ Hopf bifurcation appears near both P_2 and P_3 .
- ▶ Distinct removal rates reshape long-run ecological outcomes.

Limitations

- ▶ No empirical calibration data.
- ▶ Deterministic model only.
- ▶ No delay, noise, space, or seasonality.
- ▶ Stability conditions may be sufficient, not necessary.

Distinct removal rates are mathematically and biologically meaningful.